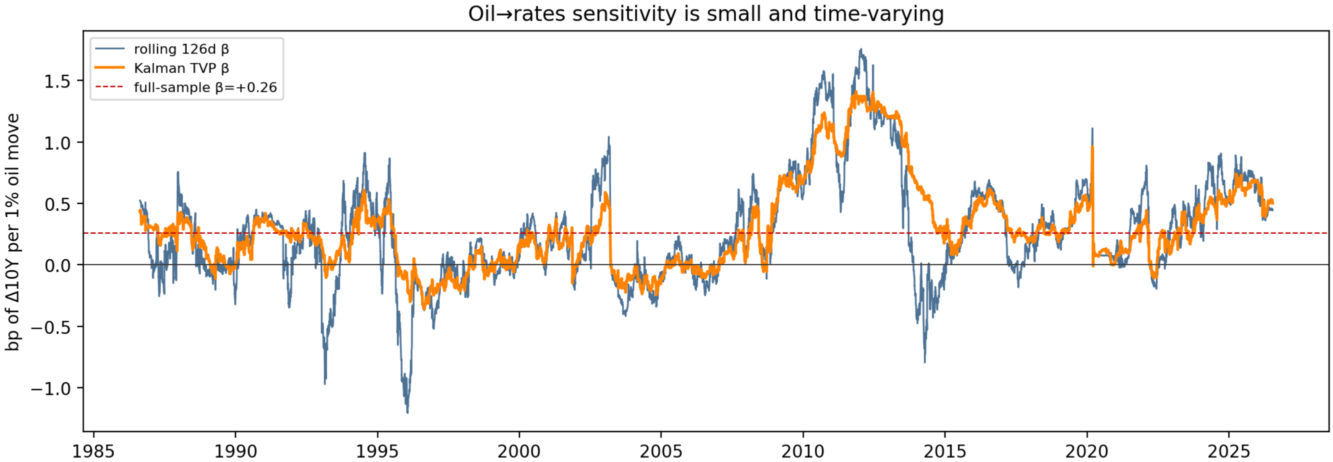
Why Don't Rates Move With Oil?

A conditional-sensitivity study: US10Y vs WTI, 1986–2026 daily (FRED, free data).

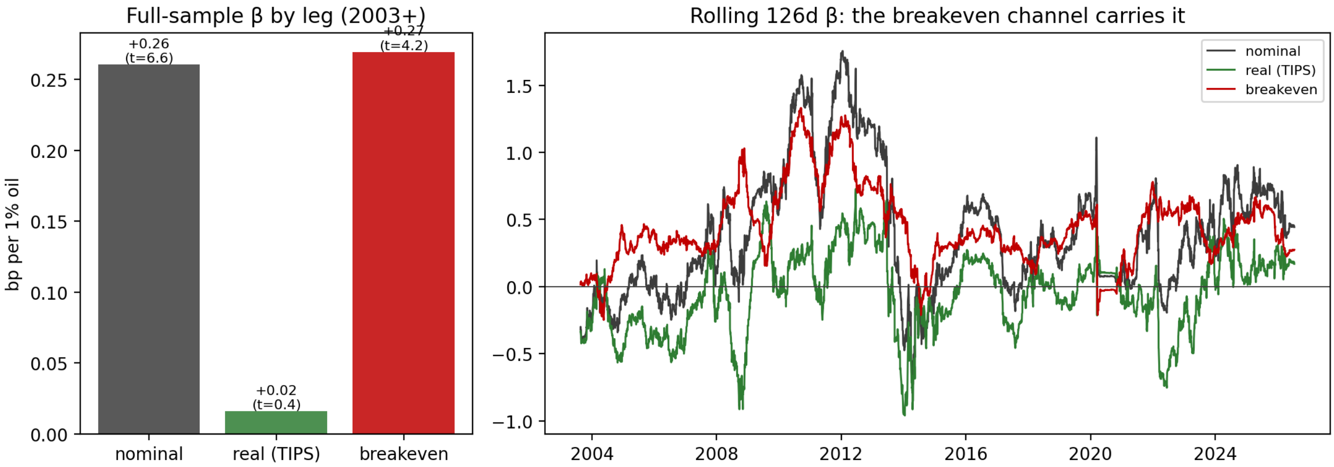
1. The puzzle, quantified

Both series are I(1); everything below runs on oil log-returns vs bp changes. The full-sample daily sensitivity is **+0.26bp per 1% oil** (HAC t=6.6) — statistically real but economically tiny: R^2 = 1.4%, so a 5% oil move maps to ≈1.3bp on the 10Y. Monthly: +0.36bp (t=2.6); Brent robustness: +0.23bp (t=6.3).



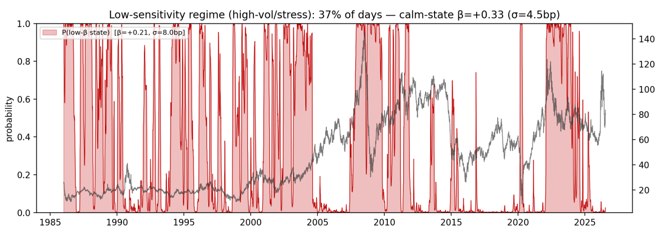
2. The mechanism: two channels that offset

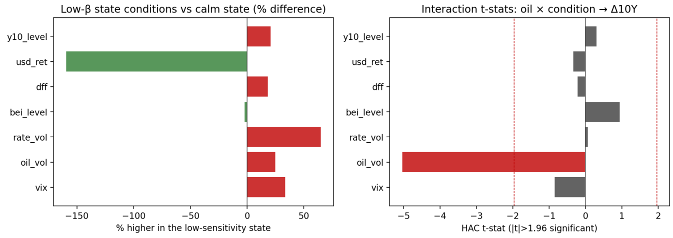
Nominal = real + breakeven. Oil's effect runs *entirely* through the inflation leg: oil→breakeven β=+0.27 (t=4.2, R^2=5.3%); oil→real β=+0.02 (t=0.4, ≈0). Granger confirms: oil→BEI p=0.004, oil→real p=0.44. On big-oil days (|ret| $>$3%, n=737) the real and breakeven legs move in **opposite directions 60% of the time** (corr -0.32) and the real leg is *larger* on average (|Δreal|=4.7bp vs |Δbei|=3.5bp) — the growth/risk channel cancels the inflation channel exactly when oil moves most.



3. The conditions for low sensitivity (regimes)

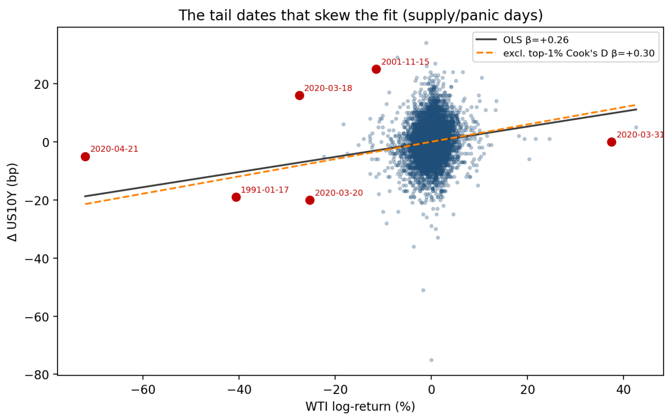
A 2-state Markov-switching regression finds a **low-sensitivity state** (β=+0.21, 37% of days) that is the *high-stress* state: VIX 23 vs 17, oil vol 2.6 vs 2.1%/d, rate vol 7.5 vs 4.5bp/d (all p$<$0.01); the calm state carries β=+0.33. The only significant interaction is **oil volatility** (t=-5.0): per-unit sensitivity *halves* as oil vol doubles — the response **saturates**, so exactly when oil moves most (supply shocks, risk-off), each 1% carries the least rate signal. In bp terms the two states respond similarly ($\beta\times$typical move ≈ 0.5 vs 0.7bp/day).

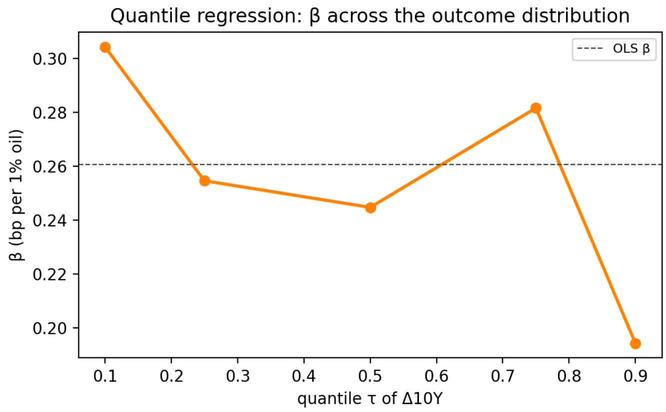




4. The datapoints that skew the distribution

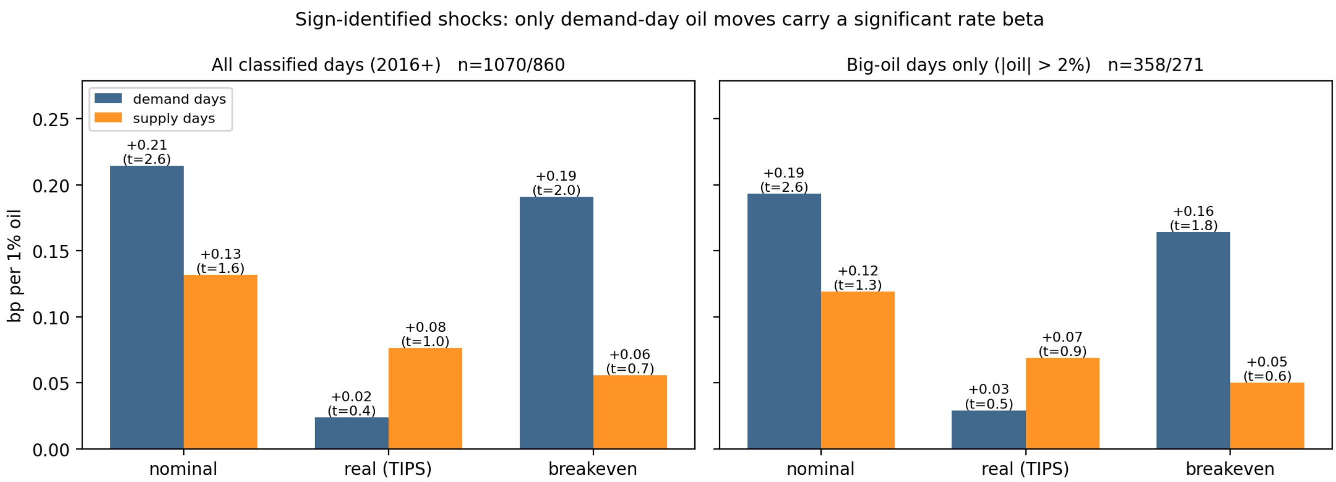
The influential tail is a handful of supply/panic days — 2020-04-21 (oil -72%, rates -5bp), the Mar-2020 COVID cluster, 17-Jan-1991 (Gulf War), Nov-2008 — where rates decoupled or moved *against* oil. They **drag the unconditional β down**: OLS +0.26 $\rightarrow$ +0.30 excluding the top-1% Cook's-D days (robust Huber +0.27). Conditional on |oil| $>$5% (n=349), the mean rate move is -0.7bp — effectively zero.





5. Demand vs supply shocks: sign identification

Classifying each day by same-day S&P 500 co-movement (oil and equities together = **demand** shock, opposite = **supply** shock; 2479 days 2016–2026, 55% demand / 45% supply — the free equity history caps the sample at ~10y): on **demand days** oil$\rightarrow$10Y β=+0.21 (t=2.6), carried by the breakeven leg (β=+0.19, t=2.0; real β=+0.02, t=0.4). On **supply days** *no* leg is significant: nominal β=+0.13 (t=1.6), breakeven β=+0.06 (t=0.7). The ordering matches the demand-shock hypothesis and holds on big-oil days (|oil| $>$2%: demand t=2.6 vs supply t=1.3) — but honestly stated, the demand–supply gap in β is *modest* (+0.21 vs +0.13bp per 1%), not dramatic: the sharper distinction is significance, not size. Most recent 21 classified days (2026-06-18–2026-07-20): 9 demand / 12 supply $\rightarrow$ a **mixed** tape, so the current oil impulse should be read with the (weaker) supply-day betas in mind.



Bottom line

Rates "don't move with oil" because (i) the pass-through runs only through breakevens and is small ($\approx$0.3bp per 1%), (ii) in stress/supply-shock regimes the real-rate (growth) leg moves opposite to the inflation leg and cancels it — and that is precisely when oil moves most, (iii) per-unit sensitivity saturates with oil volatility, and (iv) the unconditional estimate is further dragged down by a few identifiable panic days. Oil matters for *breakeven* positioning; for nominal duration it is a second-order input unless the move is demand-led in a calm regime.

Sources: FRED (DCOILWTICO, DCOILBRENTEU, DGS10, DFII10, T10YIE, VIXCLS, DTWEXBGS, DFF, SP500). WTI negative-price day 2020-04-20 excluded from returns. HAC/Newey-West errors; Markov 2-state ML; Cook's-distance influence; Kilian-style sign identification for demand/supply days. Reproduce: `python -m oil_rates.run --report`.